



AMD MI355X Can Generate Up to \$32B of Revenue per GigaWatt on Kimi K3 2.8T

Agentic serving economics of vLLM on
AMD Instinct MI355X, measured by
InferenceX AgentX

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Summary

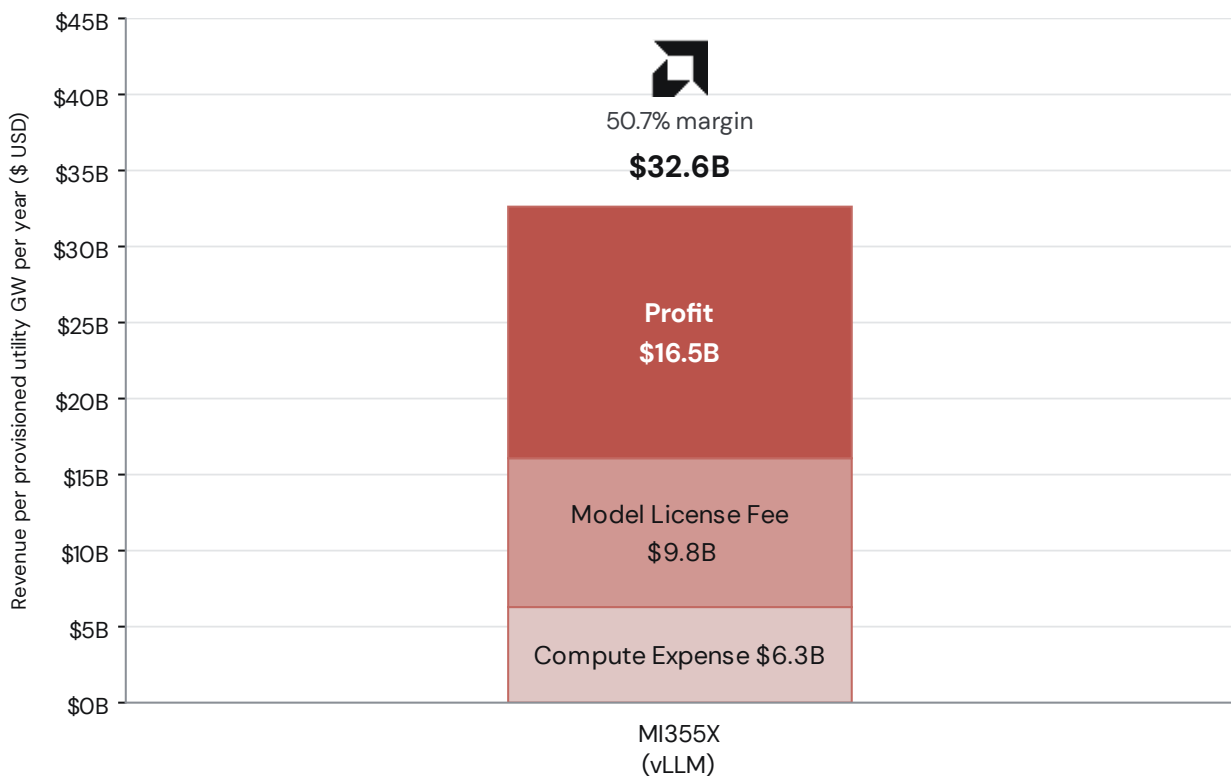
InferenceX AgentX measures how many agentic-coding tokens a production serving stack delivers at a fixed interactivity target. Priced at Kimi K3's public OpenRouter rates and scaled to one gigawatt of provisioned power, the MI355X vLLM result is **\$32.6 billion of revenue per year** at P90 45 tok/s/user and 60% utilization.

Owned at hyperscaler volume (\$1.50 per chip-hour), the fleet keeps \$16.5 billion of profit at a 50.7% margin. Rented on a 3-year contract at August 2026 pricing (\$3.00 per chip-hour), it keeps \$10.3 billion at 31.5%.

Figure 1: Kimi K3 2.8T Agentic Revenue & Profit Estimates per GigaWatt per Year at P90 45 tok/s/user

Cost Tier: Owning Hyperscaler · **TCO \$/chip/hr:** MI355X \$1.50 · **Utilization:** 60% · **Model License Fee:** 30% · **Updated:** 2026-09-04 · **Source:** SemiAnalysis InferenceX™

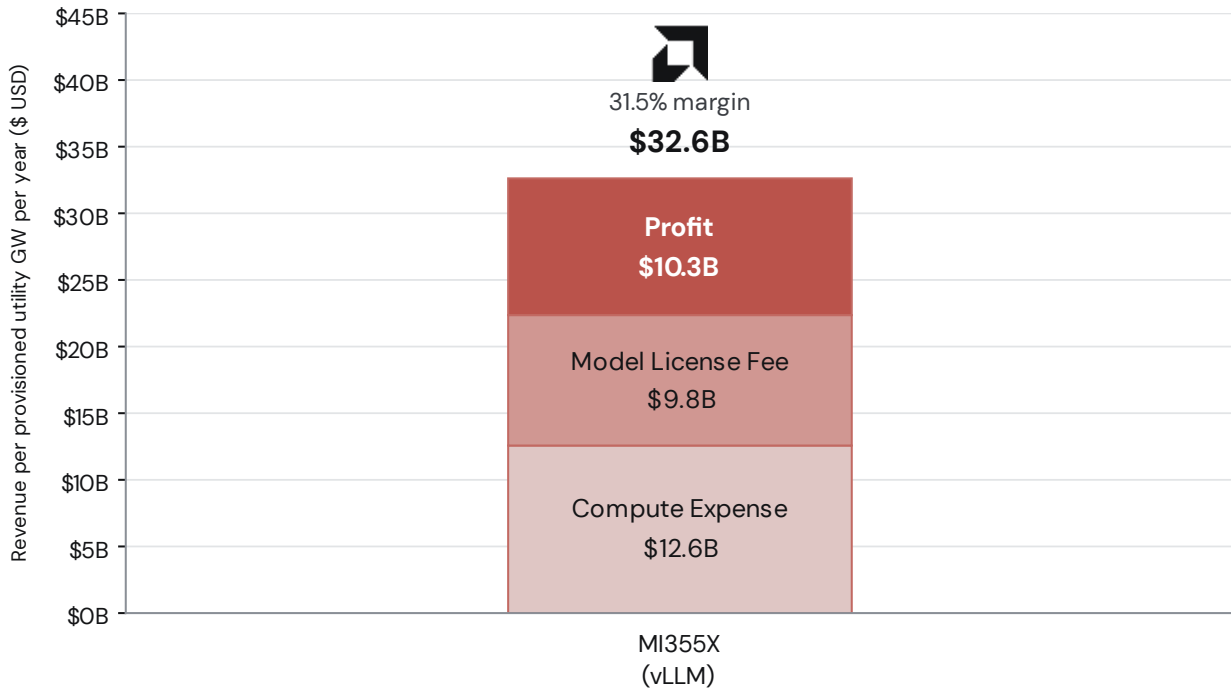
Selling Price per Million Tokens: Input \$3 · Cached Input \$0.3 · Output \$15 (OpenRouter)



Owned at hyperscaler volume. Compute is charged on every provisioned GPU-hour at \$1.50; the model lab takes 30% of revenue as a license fee; the operator keeps \$16.5B, or \$3.95 per chip-hour.

Figure 2: Kimi K3 2.8T Agentic Revenue & Profit Estimates per GigaWatt per Year at P90 45 tok/s/user

Cost Tier: 3-Year Rental (August 2026 pricing) · **TCO \$/chip/hr:** MI355X \$3.00 · **Utilization:** 60% · **Model License Fee:** 30% · **Updated:** 2026-09-04 · **Source:** SemiAnalysis InferenceX™
Selling Price per Million Tokens: Input \$3 · Cached Input \$0.3 · Output \$15 (OpenRouter)



Rented at \$3.00 per chip-hour. Revenue and license fee are unchanged; compute expense doubles to \$12.6B and profit falls to \$10.3B, or \$2.45 per chip-hour. The rented fleet breaks even at 33% utilization; the owned fleet at 16.5%.

Owned vs. rented, per gigawatt-year

Cost profile	Revenue	Compute expense	Model license fee	Profit	Margin	Profit / chip-hr
Owned, \$1.50 per chip-hour	\$32.6B	\$6.3B	\$9.8B	\$16.5B	50.7%	\$3.95
Rented, \$3.00 per chip-hour	\$32.6B	\$12.6B	\$9.8B	\$10.3B	31.5%	\$2.45

How the estimate is built

- **6,115 tok/s per GPU** at P90 45 tok/s/user, interpolated on the measured MI355X vLLM frontier (FP4, MTP, TP8).
- **478,469 GPUs per gigawatt** at 2.09 kW all-in per chip; 60% of benchmarked throughput is sold.
- **\$0.59 per million tokens** blended: 99.2% of tokens are input and 93.7% of those hit the prefix cache.
- **Profit** = revenue – compute (every GPU-hour, busy or idle) – 30% model license fee.

Sources: inferencex.semianalysis.com/profit-estimator-per-gigawatt · inferencex.semianalysis.com/agentx · openrouter.ai/moonshotai/kimi-k3 · semianalysis.com/ai-cloud-tco-model. Figures are estimates from benchmark throughput and public list prices, not forecasts of any operator's results.